

LAB EXPERIMENT RECORD

Set 1: Monthly Sales Data

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	29 September 2026
Day	Tuesday

Aim

To visualize monthly sales data using a line chart, bar chart, and scatter plot, and to build an interactive dashboard using R and Shiny.

Dataset: Monthly Sales

Month	Sales (\$)
January	15000
February	18000
March	22000
April	20000
May	23000

PROGRAM 1: Line Chart of Monthly Sales

Question

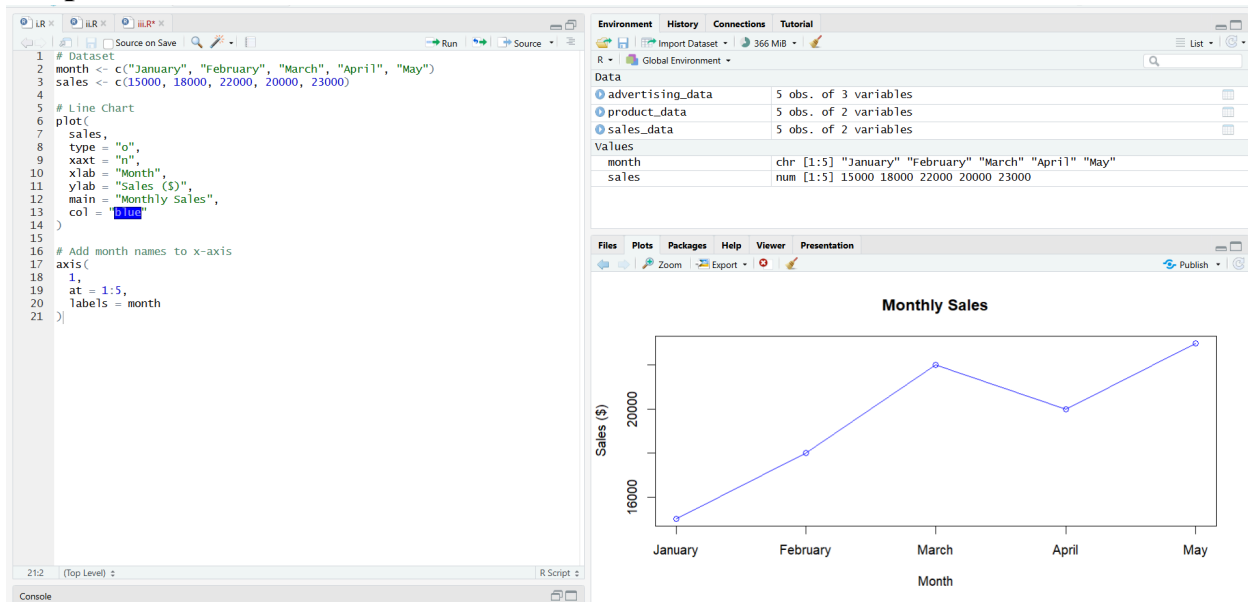
Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

R Program

```
# Dataset
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

# Line Chart
plot(
  sales,
  type = "o",
  xaxt = "n",
  xlab = "Month",
  ylab = "Sales ($)",
  main = "Monthly Sales",
  col = "blue"
)
# Add month names to x-axis
axis(
  1,
  at = 1:5,
  labels = month
)
```

Output



Observation

Sales increase from January to March, decrease slightly in April, and reach the highest value in May.

PROGRAM 2: Bar Chart of Top-Selling Products

Question

Generate a bar chart to display the top-selling products for the year. Label the chart elements.

Sample Product Data

Product	Annual Sales (\$)
Laptop	45000
Mobile	38000
Tablet	25000
Headphones	18000
Smartwatch	15000

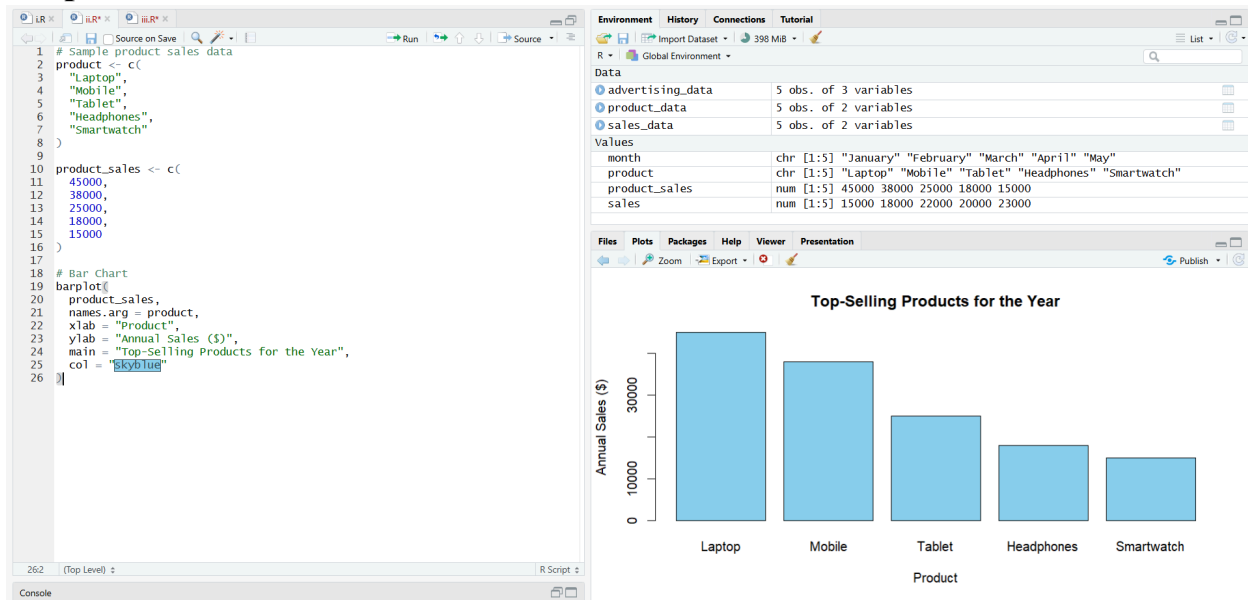
R Program

```
# Sample product sales data
product <- c(
  "Laptop",
  "Mobile",
  "Tablet",
  "Headphones",
  "Smartwatch"
)

product_sales <- c(
  45000,
  38000,
  25000,
  18000,
  15000
)

# Bar Chart
barplot(
  product_sales,
  names.arg = product,
  xlab = "Product",
  ylab = "Annual Sales ($)",
  main = "Top-Selling Products for the Year",
  col = "skyblue"
)
```

Output



Observation

The bar chart shows the sales comparison among the products. Among the sample data, Laptop has the highest annual sales, while Smartwatch has the lowest.

PROGRAM 3: Scatter Plot of Advertising Budget vs Monthly Sales

Question

Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.

Sample Advertising Data

Month	Advertising Budget (\$)	Sales (\$)
January	5000	15000
February	6000	18000
March	7000	22000
April	8000	20000
May	9000	23000

R Program

```
# Monthly sales data
sales <- c(
  15000,
  18000,
  22000,
  20000,
  23000
)
# Sample advertising budget data
advertising <- c(
  5000,
  6000,
  7000,
  8000,
  9000
)
# Scatter Plot
plot(
  advertising,
  sales,
  main = "Advertising Budget vs Monthly Sales",
  xlab = "Advertising Budget ($)",
```

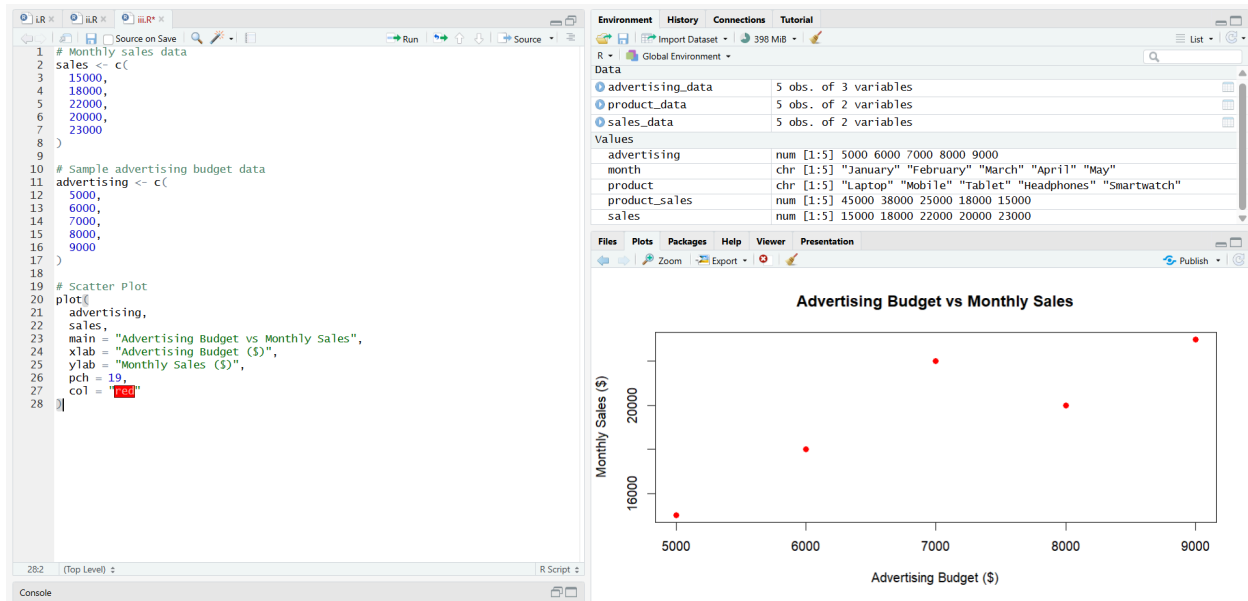
```

      ylab = "Monthly Sales ($)",
      pch = 19,
      col = "red"
    )

```

Output

Advertising Budget vs Monthly Sales Scatter Plot



Insight

The scatter plot shows a generally positive relationship between advertising budget and monthly sales. As the advertising budget increases, monthly sales generally tend to increase. This indicates that higher advertising spending is associated with higher sales for the sample values used.

PROGRAM 4: Interactive Sales Dashboard

Question

Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

R Program

```
# Load Shiny
library(shiny)

# -----
# Monthly Sales Data
# -----

month <- c(
  "January",
  "February",
  "March",
  "April",
  "May"
)

sales <- c(
  15000,
  18000,
  22000,
  20000,
  23000
)

# -----
# Product Sales Data
# -----

product <- c(
  "Laptop",
  "Mobile",
  "Tablet",
  "Headphones",
  "Smartwatch"
)

product_sales <- c(
  45000,
  38000,
  25000,
```

```

    18000,
    15000
  )
# -----
# User Interface
# -----
ui <- fluidPage(
  titlePanel("Monthly Sales Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput(
        "chart",
        "Select Chart:",
        choices = c(
          "Line Chart",
          "Bar Chart"
        )
      )
    )
  ),

  mainPanel(
    plotOutput("salesPlot")

  )
)

# -----
# Server
# -----

server <- function(input, output) {

  output$salesPlot <- renderPlot({

    if (input$chart == "Line Chart") {

      # Line Chart
      plot(
        sales,

```

```

    type = "o",
    xaxt = "n",
    xlab = "Month",
    ylab = "Sales ($)",
    main = "Monthly Sales",
    col = "blue"
  )

  axis(
    1,
    at = 1:5,
    labels = month
  )

} else {

  # Bar Chart
  barplot(
    product_sales,
    names.arg = product,
    xlab = "Product",
    ylab = "Annual Sales ($)",
    main = "Top-Selling Products",
    col = "skyblue"
  )

}

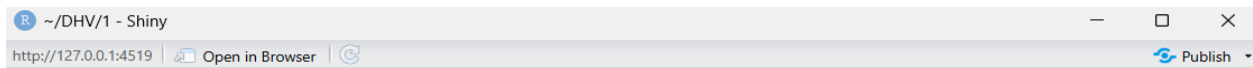
})
}

# -----
# Run Dashboard
# -----

shinyApp(
  ui = ui,
  server = server
)

```

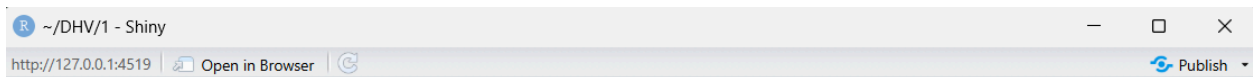
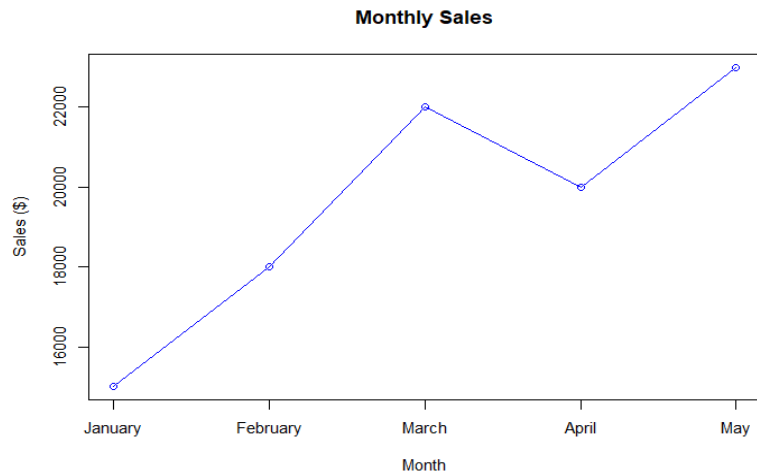
Dashboard Output



Monthly Sales Dashboard

Select Chart:

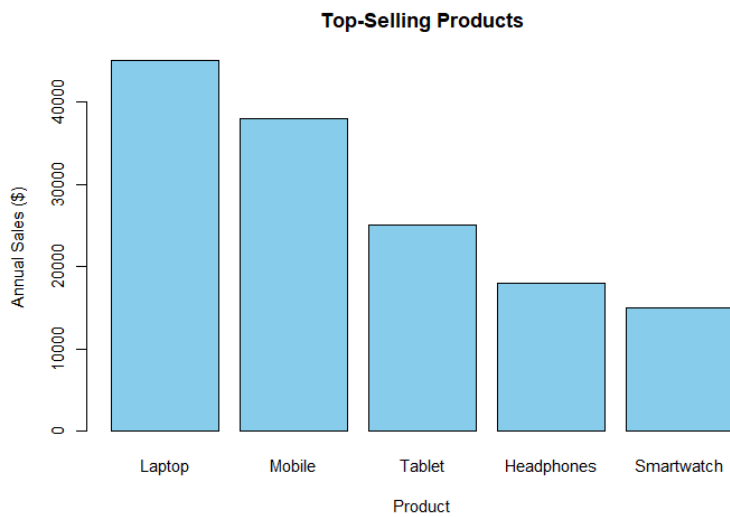
Line Chart



Monthly Sales Dashboard

Select Chart:

Bar Chart



Observation

The interactive dashboard provides a chart-selection control that allows users to switch between the monthly sales line chart and the top-selling products bar chart.

Procedure

1. Create the monthly sales vectors in R.
2. Use `plot()` to create the monthly sales line chart.
3. Add month names to the x-axis using `axis()`.
4. Create the product and annual sales vectors for the bar chart.
5. Use `barplot()` to visualize the top-selling products.
6. Create advertising-budget and monthly-sales vectors of equal length.
7. Use `plot()` to create the scatter plot.
8. Observe the relationship between advertising budget and sales.
9. Install and load the `shiny` package.
10. Create the Shiny UI and server components.
11. Run the Shiny application.
12. Use the chart-selection box to switch between the line chart and bar chart.

Result

The monthly sales data was successfully visualized using a line chart, the top-selling products were represented using a bar chart, and the relationship between advertising budget and monthly sales was explored using a scatter plot. An interactive Shiny dashboard was also created to allow users to switch between the line chart and bar chart.

SET 2: CUSTOMER FEEDBACK ANALYSIS

Aim

To analyze customer satisfaction data using a histogram, pie chart, stacked bar chart, and word cloud to understand customer age distribution, satisfaction scores, age-group-wise satisfaction, and common feedback terms.

Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

PROGRAM 1: Histogram of Customer Ages

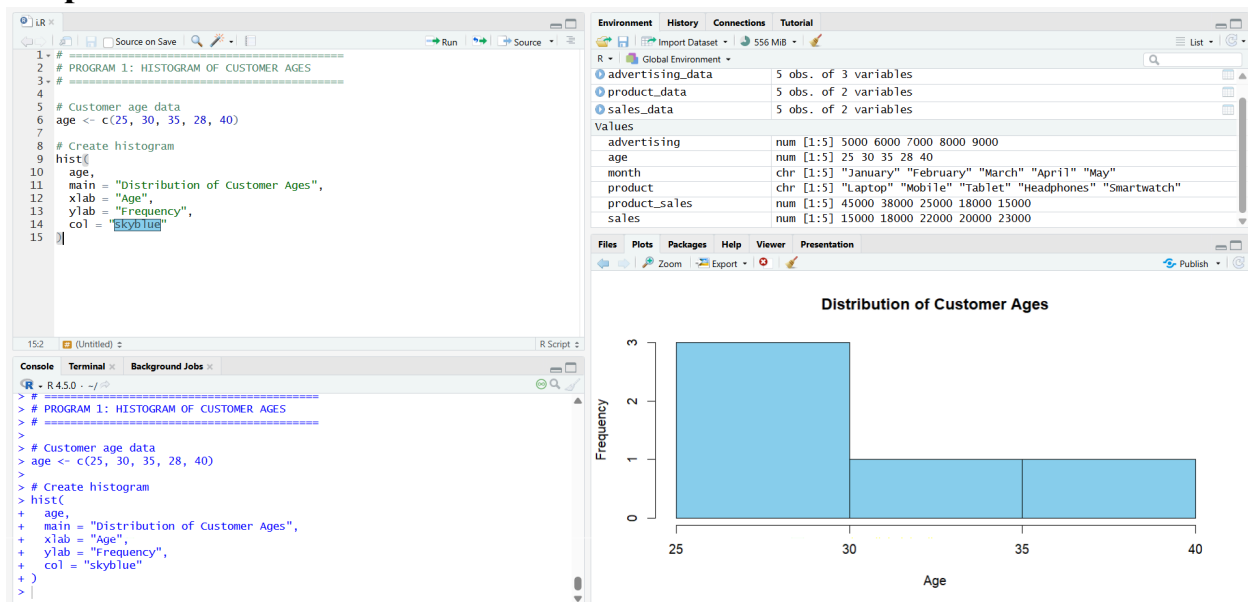
Question

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

R Program

```
# =====  
# PROGRAM 1: HISTOGRAM OF CUSTOMER AGES  
# =====  
# Customer age data  
age <- c(25, 30, 35, 28, 40)  
# Create histogram  
hist(  
  age,  
  main = "Distribution of Customer Ages",  
  xlab = "Age",  
  ylab = "Frequency",  
  col = "skyblue"  
)
```

Output



Observation

The histogram represents the distribution of customer ages in the given dataset.

PROGRAM 2: Pie Chart of Satisfaction Scores

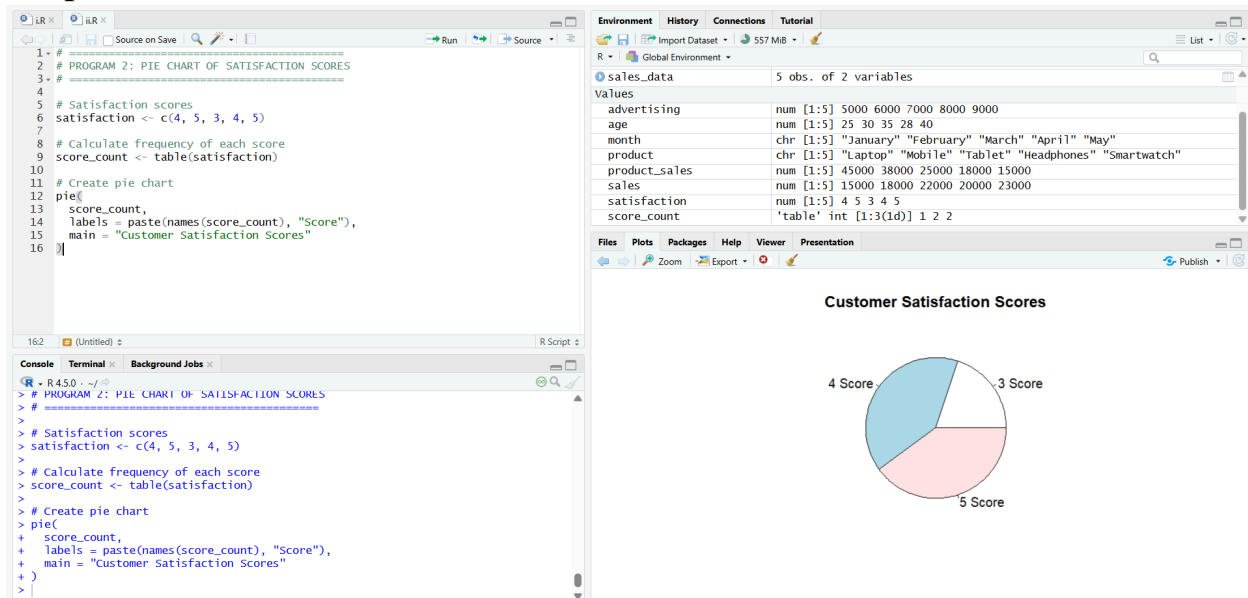
Question

Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

R Program

```
# =====  
# PROGRAM 2: PIE CHART OF SATISFACTION SCORES  
# =====  
# Satisfaction scores  
satisfaction <- c(4, 5, 3, 4, 5)  
# Calculate frequency of each score  
score_count <- table(satisfaction)  
# Create pie chart  
pie(  
  score_count,  
  labels = paste(names(score_count), "Score"),  
  main = "Customer Satisfaction Scores"  
)
```

Output



Observation

Satisfaction scores 4 and 5 each occur twice, while satisfaction score 3 occurs once.

PROGRAM 3: Stacked Bar Chart by Age Group

Question

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

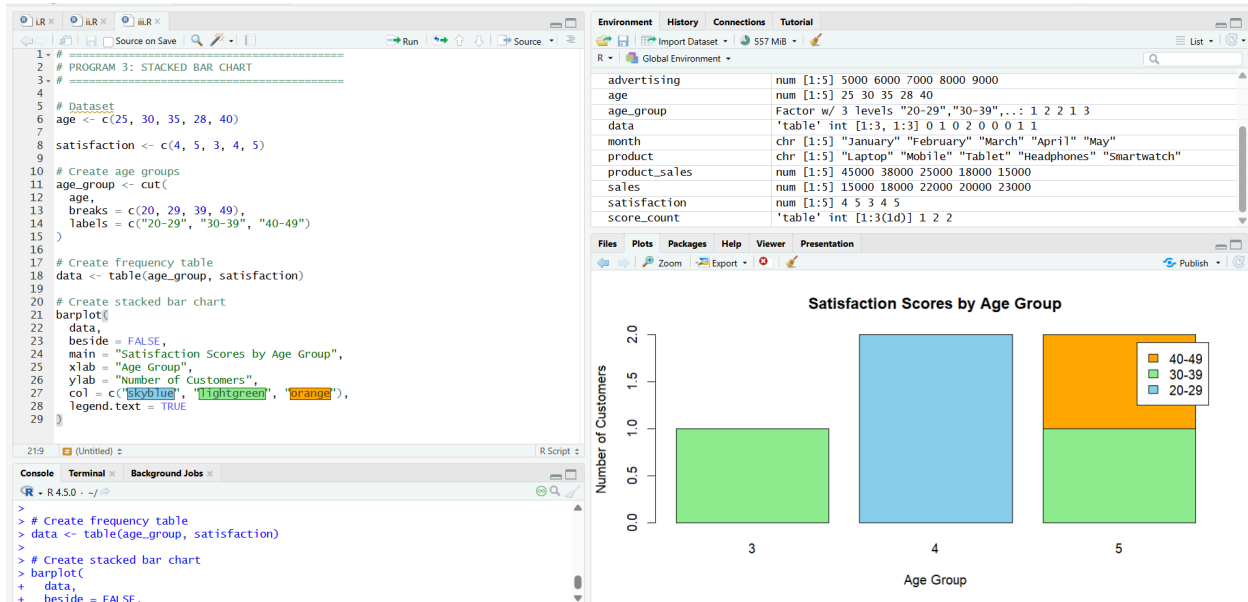
We first divide the customers into three age groups:

- 20–29
- 30–39
- 40–49

R Program

```
# =====  
# PROGRAM 3: STACKED BAR CHART  
# =====  
  
# Dataset  
age <- c(25, 30, 35, 28, 40)  
  
satisfaction <- c(4, 5, 3, 4, 5)  
  
# Create age groups  
age_group <- cut(  
  age,  
  breaks = c(20, 29, 39, 49),  
  labels = c("20-29", "30-39", "40-49")  
)  
  
# Create frequency table  
data <- table(age_group, satisfaction)  
  
# Create stacked bar chart  
barplot(  
  data,  
  beside = FALSE,  
  main = "Satisfaction Scores by Age Group",  
  xlab = "Age Group",  
  ylab = "Number of Customers",  
  col = c("skyblue", "lightgreen", "orange"),  
  legend.text = TRUE  
)
```

Output



Observation

The stacked bar chart represents the distribution of satisfaction scores across the three age groups.

PROGRAM 4: Word Cloud of Customer Feedback

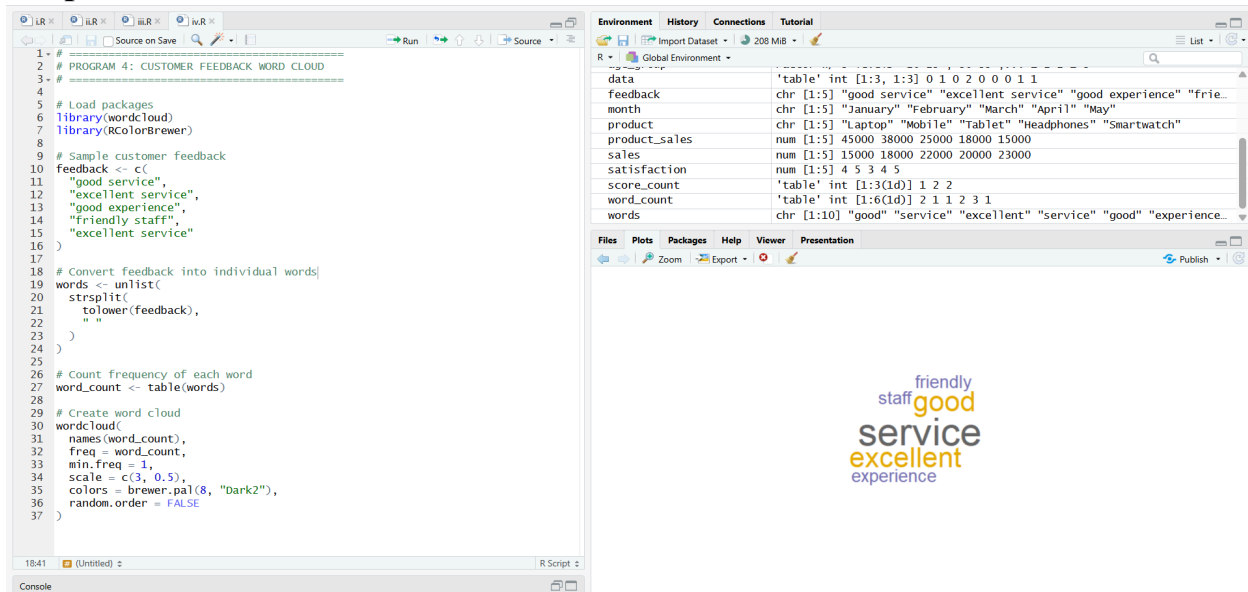
Question

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

R Program

```
# =====  
# PROGRAM 4: CUSTOMER FEEDBACK WORD CLOUD  
# =====  
  
# Load packages  
library(wordcloud)  
library(RColorBrewer)  
# Sample customer feedback  
feedback <- c(  
  "good service",  
  "excellent service",  
  "good experience",  
  "friendly staff",  
  "excellent service"  
)  
# Convert feedback into individual words  
words <- unlist(  
  strsplit(  
    tolower(feedback),  
    " "  
  )  
)  
# Count frequency of each word  
word_count <- table(words)  
  
# Create word cloud  
wordcloud(  
  names(word_count),  
  freq = word_count,  
  min.freq = 1,  
  scale = c(3, 0.5),  
  colors = brewer.pal(8, "Dark2"),  
  random.order = FALSE  
)
```

Output



Observation

The word cloud highlights frequently occurring terms in the supplied sample feedback, including service, excellent, good, friendly, and experience.

Procedure

1. Create the customer age and satisfaction score vectors in R.
2. Use `hist()` to represent the distribution of customer ages.
3. Use `table()` and `pie()` to display the satisfaction score distribution.
4. Create age groups using `cut()`.
5. Create a frequency table using `table()`.
6. Use `barplot()` to create the stacked bar chart.
7. Install and load the `wordcloud` and `RColorBrewer` packages.
8. Prepare sample open-ended customer feedback.
9. Split the feedback into individual words.
10. Count the frequency of each word.
11. Use `wordcloud()` to visualize the frequently occurring words.

Result

The customer satisfaction data was successfully visualized using a histogram, pie chart, and stacked bar chart. A word cloud was also created from sample customer feedback to identify frequently occurring feedback terms.

SET 3: EMPLOYEE PERFORMANCE EVALUATION

Aim

To analyze employee performance using line, bar, and scatter charts, and to develop an interactive Tableau dashboard for exploring employee performance and department-wise distribution.

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

PROGRAM 1: Line Chart of Employee Performance

Question

Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

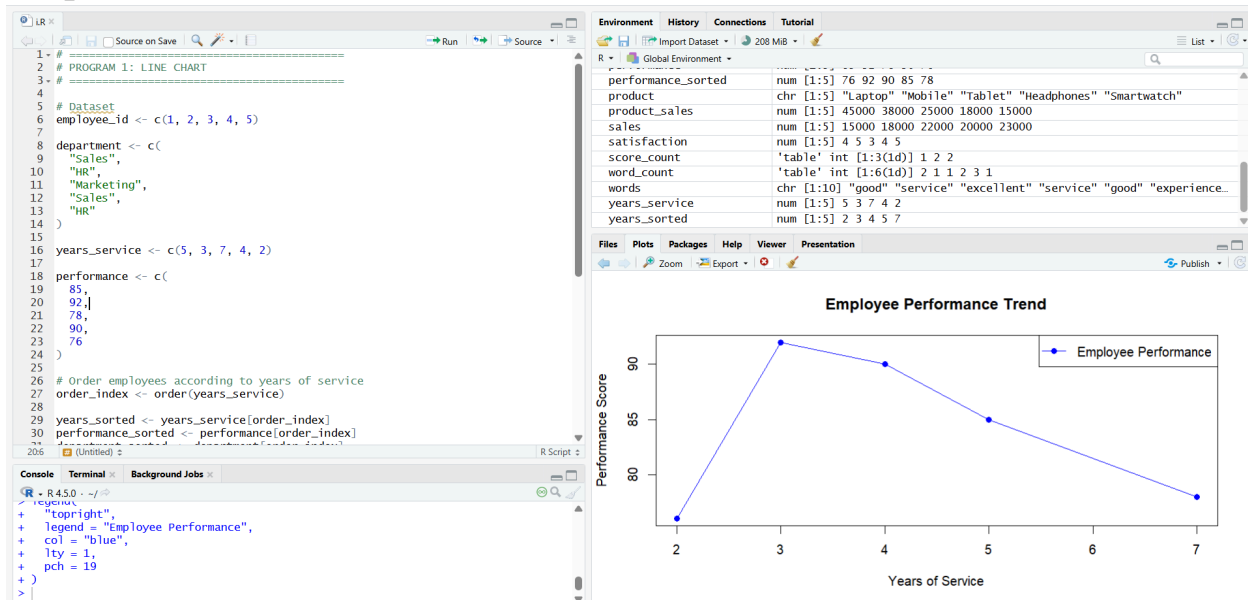
Since the dataset doesn't contain dates or years, **Years of Service** is used as the time/progression variable.

R Program

```
# =====  
# PROGRAM 1: LINE CHART  
# =====  
# Dataset  
employee_id <- c(1, 2, 3, 4, 5)  
department <- c(  
  "Sales",  
  "HR",  
  "Marketing",  
  "Sales",  
  "HR"  
)  
years_service <- c(5, 3, 7, 4, 2)  
performance <- c(  
  85,  
  92,  
  78,  
  90,  
  76  
)  
# Order employees according to years of service  
order_index <- order(years_service)  
years_sorted <- years_service[order_index]  
performance_sorted <- performance[order_index]  
department_sorted <- department[order_index]  
# Create line chart  
plot(  
  years_sorted,  
  performance_sorted,  
  type = "o",  
  pch = 19,  
  xlab = "Years of Service",  
  ylab = "Performance Score",  
  main = "Employee Performance Trend",  
  col = "blue"
```

```
)
# Add legend
legend(
  "topright",
  legend = "Employee Performance",
  col = "blue",
  lty = 1,
  pch = 19
)
```

Output



Observation

The performance scores vary across employees with different years of service. The highest performance score is 92 for the employee with 3 years of service, while the lowest score is 76 for the employee with 2 years of service.

PROGRAM 2: Bar Chart of Employees by Department

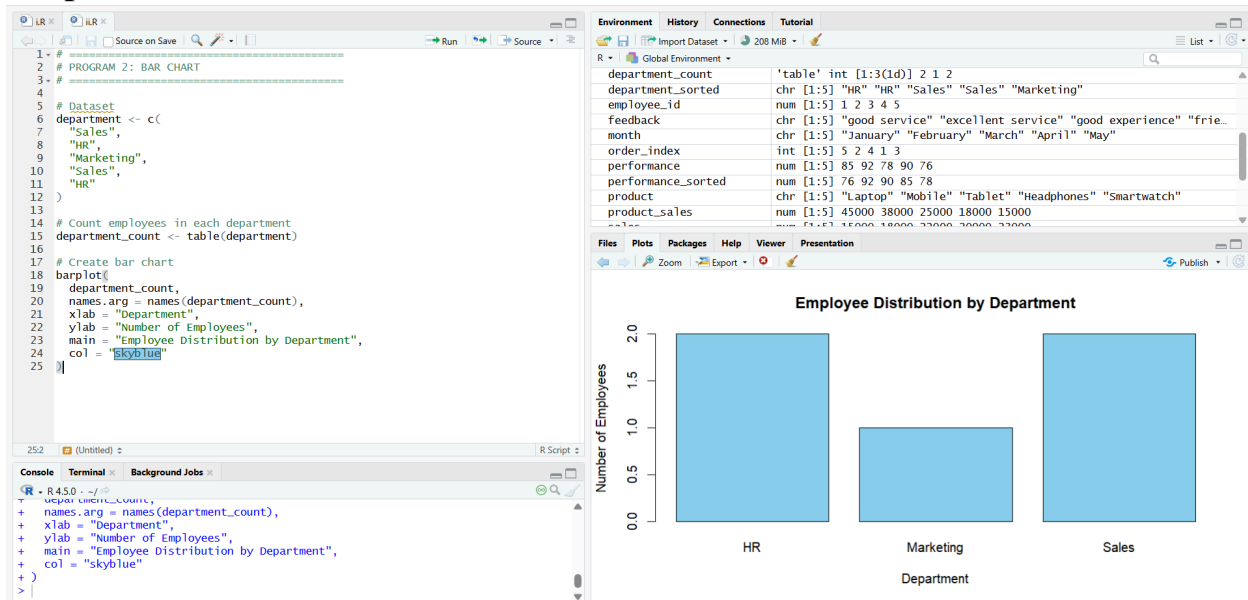
Question

Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

R Program

```
# =====  
# PROGRAM 2: BAR CHART  
# =====  
  
# Dataset  
department <- c(  
  "Sales",  
  "HR",  
  "Marketing",  
  "Sales",  
  "HR"  
)  
  
# Count employees in each department  
department_count <- table(department)  
  
# Create bar chart  
barplot(  
  department_count,  
  names.arg = names(department_count),  
  xlab = "Department",  
  ylab = "Number of Employees",  
  main = "Employee Distribution by Department",  
  col = "skyblue"  
)
```

Output



Observation

Sales and HR have the highest number of employees with two employees each, while Marketing has one employee.

PROGRAM 3: Scatter Plot — Years of Service vs Performance Score

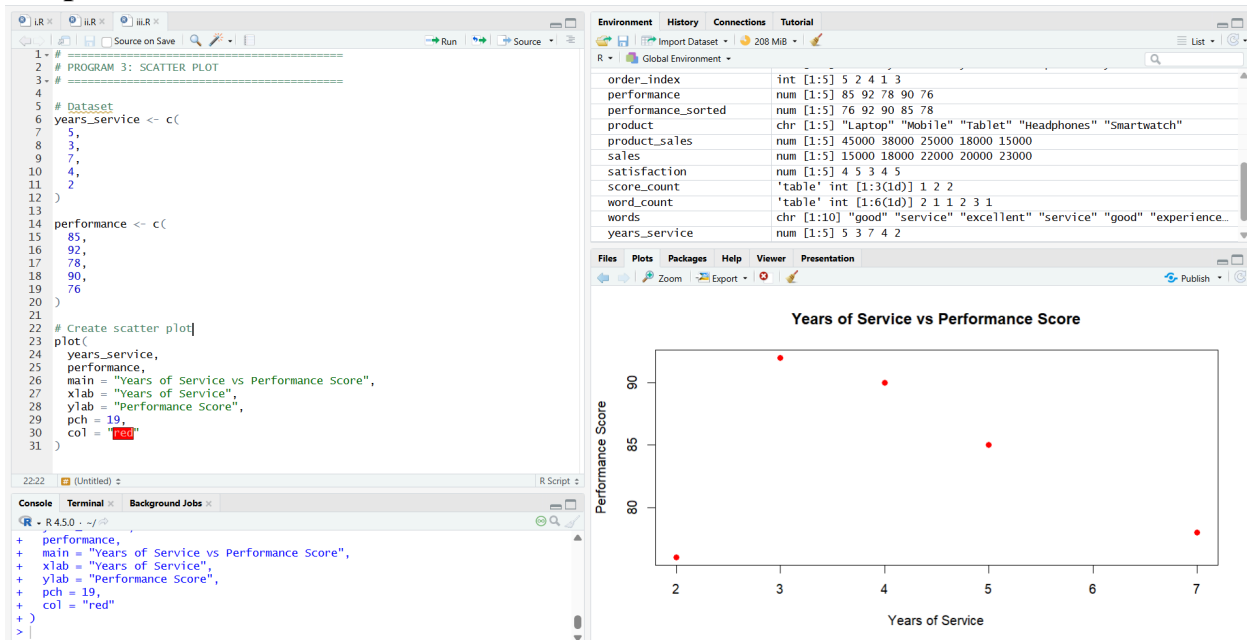
Question

Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

R Program

```
# =====  
# PROGRAM 3: SCATTER PLOT  
# =====  
  
# Dataset  
years_service <- c(  
  5,  
  3,  
  7,  
  4,  
  2  
)  
  
performance <- c(  
  85,  
  92,  
  78,  
  90,  
  76  
)  
  
# Create scatter plot  
plot(  
  years_service,  
  performance,  
  main = "Years of Service vs Performance Score",  
  xlab = "Years of Service",  
  ylab = "Performance Score",  
  pch = 19,  
  col = "red"  
)
```

Output



Insight

The scatter plot shows that performance scores vary across different years of service. In this small dataset, employees with more years of service do not consistently have higher performance scores. Therefore, years of service alone does not show a clear positive relationship with performance score in the given data.

PROGRAM 4: Tableau Dashboard

Question

Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

Step 1 — Prepare your data

Create an Excel file with the following data:

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Save it as:

Employee_Performance.xlsx

Step 2 — Open Tableau

Open **Tableau Desktop** or **Tableau Public**.

Choose:

Connect → Microsoft Excel

Then select:

Employee_Performance.xlsx

Step 3 — Create Line Chart

Go to **Sheet 1**.

Drag:

Years of Service



Columns

Drag:

Performance Score



Rows

Change the Marks type to:

Line

Drag:

Department

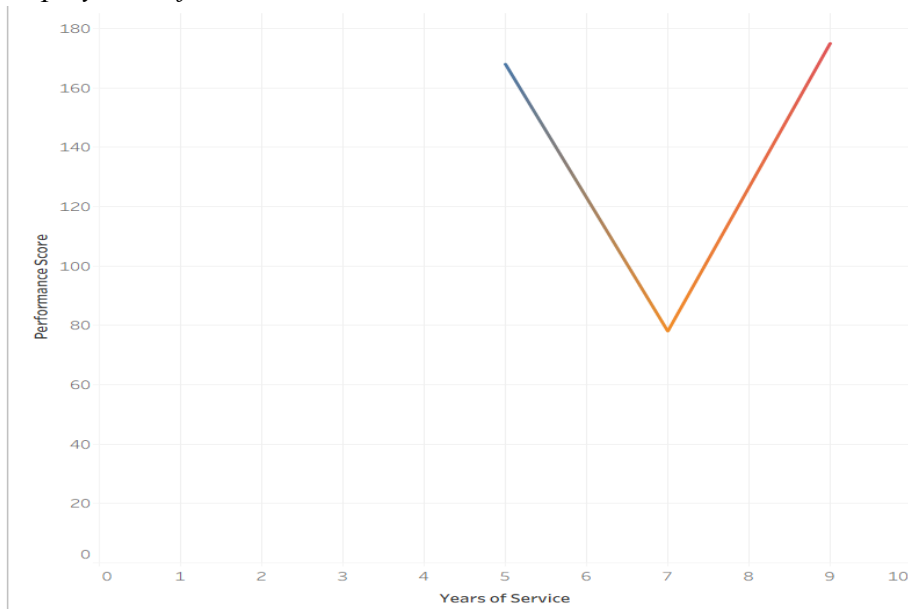


Color

This gives you a line chart with department information represented by the legend.

Chart title

Employee Performance Trend



Step 4 — Create Bar Chart

Create another worksheet.

Drag:

Department



Columns

Then use:

Employee ID



Rows

Change the aggregation of Employee ID to:

COUNT

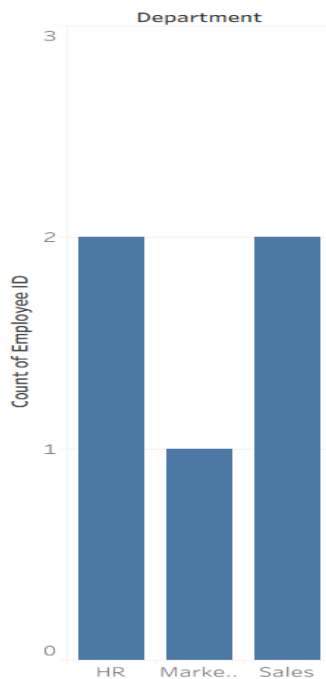
or use **Number of Records**.

Set Marks to:

Bar

Chart title

Employee Distribution by Department



The resulting chart should show:

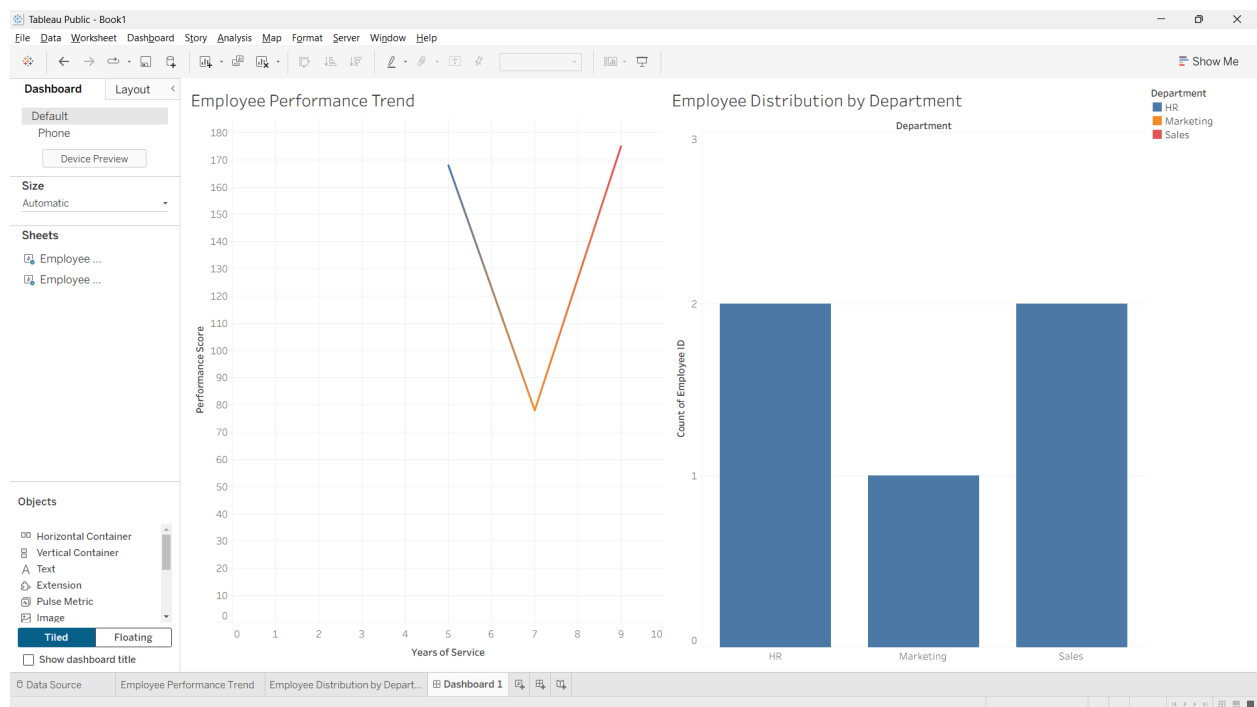
Sales 2
HR 2
Marketing 1

Step 5 — Create Dashboard

Click:

New Dashboard

Add both worksheets:



Observation

The Tableau dashboard combines the employee performance trend and department-wise employee distribution. Users can interact with the dashboard by selecting departments or applying available filters.

Procedure

1. Create the employee performance dataset.
2. Use Years of Service as the progression variable for the performance trend.

3. Create a line chart using Years of Service and Performance Score.
4. Add a legend to identify the employee performance series.
5. Count the employees in each department.
6. Use `barplot()` to create the department-wise employee distribution.
7. Use `plot()` to create the scatter plot between Years of Service and Performance Score.
8. Calculate the correlation between Years of Service and Performance Score.
9. Import the Employee Performance dataset into Tableau.
10. Create a line chart in Tableau.
11. Create a department-wise bar chart.
12. Combine both charts into an interactive Tableau dashboard.

Result

The employee performance dataset was successfully visualized using a line chart, department-wise bar chart, and scatter plot. The relationship between years of service and performance score was analyzed. An interactive Tableau dashboard was also created by combining the performance trend and department distribution charts.